

Terminal

Infrastructure: Power Grid

Budget: \$5,00,000

Location: Dubai, UAE

Target Year: 2040

Risk Level: LOW

Risk Score: 0.48/100

Climate Resilience and Engineering Report for Dubai Power Grid

The report focuses on enhancing the resilience of the power grid infrastructure in Dubai, given the coordinates [25.6649, 55.8230], [25.6640, 55.8244], [25.6629, 55.8231], [25.6642, 55.8219], against flood hazards by 2040.

STRATEGY 1

Elevating Critical Infrastructure Components: Raising the level of critical power grid components such as substations, transformers, and switchgear above the projected flood levels will prevent water ingress and minimize the risk of electrical failures. This can be achieved through the construction of elevated platforms or by relocating these components to higher ground. The design should consider the projected 0.7m sea level rise and intensified rainfall events.

STRATEGY 2

Implementing Flood-Resistant Materials and Designs: Utilizing flood-resistant materials for the construction and retrofitting of power grid infrastructure can significantly enhance resilience. This includes the use of waterproof coatings, stainless steel, and fiber-reinforced polymers for structural components. Additionally, designing the infrastructure with flood-resilient features such as watertight enclosures and buoyant foundations can help maintain functionality during flood events.

STRATEGY 3

Enhancing Drainage and Water Management Systems: Integrating advanced drainage systems around the power grid infrastructure can mitigate the impact of flooding. This involves designing and installing efficient stormwater drainage networks that can quickly redirect floodwaters away from critical components. The system should be capable of handling the increased rainfall intensity projected due to climate change, leveraging insights from the Dh30 billion 'Tasreef' drainage project.

RESILIENCE SCORE: 92%

